Differential Equations Workbook

Partial differential equations

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MATH

SEPARATION OF VARIABLES

■ 1. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$\frac{1}{k^2}\left(\frac{\partial^2 u}{\partial x^2}\right)+\frac{\partial u}{\partial t}=0$$

■ 2. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$16u_{yy}+u_{xx}=0$$

■ 3. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$9\frac{\partial^2 u}{\partial x^2}-4\frac{\partial^2 u}{\partial y^2}=0$$

■ 4. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$-\frac{\partial^2 u}{\partial x^2}+25\frac{\partial^2 u}{\partial t^2}=0$$

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■ 5. Solve the the partial differential equation by separating variables and assuming the separation constant is positive.

$$-\frac{\partial^2 u}{\partial x^2}+\frac{\partial u}{\partial t}=0$$

■ 6. Solve the the partial differential equation by separating variables and assuming the separation constant is negative.

$$\frac{\partial u}{\partial x}+\frac{\partial^2 u}{\partial y^2}=0$$

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BOUNDARY VALUE PROBLEMS

■ 1. Solve the boundary value problem, if $y(0) = 3$ and $y'(0) = -4$.

$$y'' + 4y = 0$$

■ 2. Solve the boundary value problem, if $y(0) = 1$, $y'(0) = 0$, and $y''(0) = 5$.

$$y''' - y'' + 9y' - 9y = 0$$

■ 3. Solve the boundary value problem, if $u(\pi/3,t) = 2e^{3t}$, $(\partial u/\partial x)(\pi/3,t) = -3e^{3t}$, and $-\lambda = -9$.

$$\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial u}{\partial t} = 0$$

■ 4. Solve the boundary value problem, if $u(\pi, t) = 0$, $u(x,0) = -2\cos(x/2)$, $(\partial u/\partial x)(\pi, t) = 2e^{-t} - e^{t}$, $(\partial u/\partial t)(x,0) = 6\cos(x/2)$, and $-\lambda = -\frac{1}{4}$.

$$4\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial t^2} = 0$$

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■ 5. Solve the boundary value problem, if $u(0,y) = e^{8y} + 3e^{-8y}$, $u(x,0) = 8e^{2x} - 4e^{-2x}$, $(\partial u/\partial x)(0,y) = 6e^{8y} + 18e^{-8y}$, $(\partial u/\partial y)(x,0) = 16e^{-2x} - 32e^{2x}$, and $-\lambda = 4$.

$$\frac{\partial^2 u}{\partial y^2} - 16\frac{\partial^2 u}{\partial x^2} = 0$$

■ 6. Solve the boundary value problem, if $u = (0,y) = 4\sin(\sqrt{3\lambda}y)$.

$$3\frac{\partial u}{\partial x} - \frac{\partial^2 u}{\partial y^2} = 0$$

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THE HEAT EQUATION

■ 1. Find a solution to the partial differential equation if $u(x,0) = \sin(\pi x) + 3\sin(\pi x/2)$ and $u(0,t) = u(6,t) = 0$.

$$\frac{\partial u}{\partial t} = 9\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 6$$

■ 2. Find a solution to the partial differential equation if $u(x,0) = 2\cos(\pi x/2L)$, $u_x(0,t) = 0$, and $u(L,t) = 0$.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le L$$

■ 3. Find a solution to the partial differential equation if $u(x,0) = 4\sin(\pi x/6)\cos^2(\pi x/6)$, $u(0,t) = 0$, and $u_x(3,t) = 0$.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 3$$

■ 4. Find a solution to the partial differential equation if $u(x,0) = 2x + 5$, $u_x(0,t) = 0$, and $u_x(L,t) = 0$.

$$\frac{\partial u}{\partial t} = 25\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le L$$

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■ 5. Find a solution to the partial differential equation if $u(x,0) = x \sin x$, $u(0,t) = 0$, and $u(\pi, t) = 0$.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq \pi$$

■ 6. Find a solution to the partial differential equation if $u(x,0) = \sin(\pi x)$, $u_x(0,t) = 0$, and $u_x(5,t) = 0$.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \leq x \leq 5$$

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CHANGING THE TEMPERATURE BOUNDARIES

■ 1. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le \pi$$

$$u(x,0) = 12x + \pi$$

$$u(0,t) = \pi \qquad u(\pi, t) = 3\pi$$

■ 2. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 4$$

$$u(x,0) = x^2 + 1$$

$$u(0,t) = 1 \qquad u(4,t) = 17$$

■ 3. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 3\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 1$$

$$u(x,0) = x + \sin(\pi x)$$

$$u(0,t) = 4 \qquad u(1,t) = 5$$

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■ 4. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 5\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le 10$$

$$u(x,0) = 6 - 2x$$

$$u(0,t) = 10 \qquad u(10,t) = 50$$

■ 5. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le \frac{\pi}{2}$$

$$u(x,0) = \cos x$$

$$u(0,t) = 1 \qquad u\left(\frac{\pi}{2}, t\right) = 0$$

■ 6. Solve the partial differential equation.

$$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2} \qquad 0 \le x \le \pi$$

$$u(x,0) = e^x + x + 1$$

$$u(0,t) = 1 \qquad u(\pi, t) = 1 + \pi$$

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LAPLACE'S EQUATION

■ 1. Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq \pi$ and $0 \leq y \leq 1$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 3\sin(2x) \qquad u(0,y) = 0$$

$$u(x,1) = 0 \qquad u(\pi, y) = 0$$

■ 2. Use the product solution to find $u(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 3$ and $0 \leq y \leq 3$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u(0,y) = 0$$

$$u(x,3) = 2\sin(\pi x) + \sin(2\pi x) \qquad u(3,y) = 0$$

■ 3. Use the product solution to find $u(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq \pi$ and $0 \leq y \leq 2\pi$.

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$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u_y(x,0) = 0 \qquad u(0,y) = 2 + \cos(2y)$$

$$u_y(x,2\pi) = 0 \qquad u(\pi, y) = 0$$

■ 4. Use the product solution to find $u(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 4$ and $0 \leq y \leq 6$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u_x(0,y) = 0$$

$$u_y(x,6) = 0 \qquad u(4,y) = \sin\left(\frac{\pi y}{4}\right) + 5\sin\left(\frac{\pi y}{12}\right)$$

■ 5. Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = x \qquad u(0,y) = 0$$

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$$u(x,1) = 0 \qquad u(1,y) = \sin(3\pi y)$$

■ 6. Use the product solution to find $u_1(x, y)$, the solution to Laplace's equation along the bottom edge of the rectangle defined on $0 \leq x \leq 2\pi$ and $0 \leq y \leq 2\pi$.

$$\frac{\partial^2 u_2}{\partial x^2} + \frac{\partial^2 u_2}{\partial y^2} = 0$$

$$u(x,0) = \cos x \qquad u_x(0,y) = 0$$

$$u(x,2\pi) = \cos x \qquad u_x(2\pi, y) = 0$$

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